



DCA0305

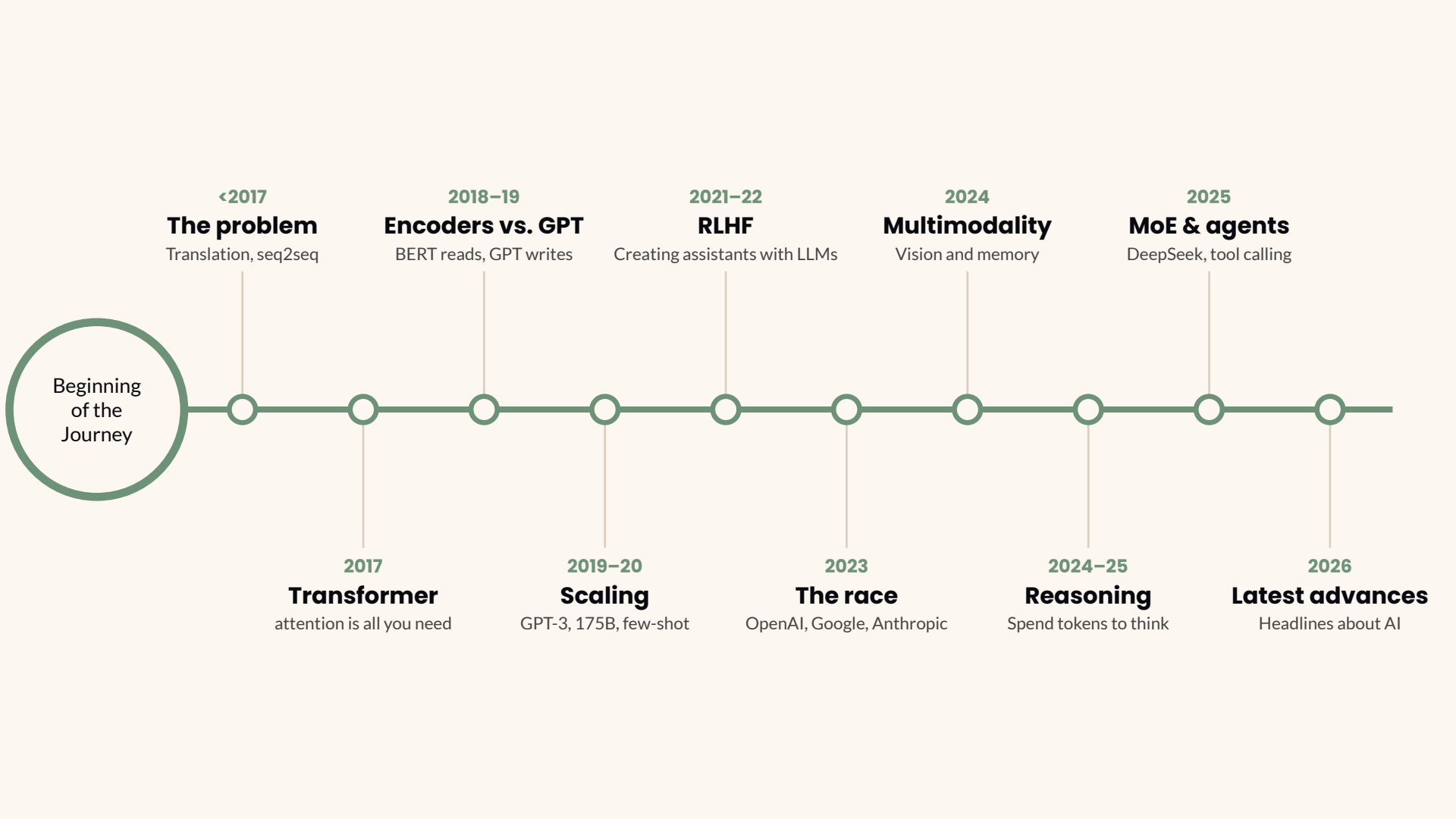
morsinaldo.medeiros.075@ufrn.edu.br
ivanovitch.silva@ufrn.br

A Brief History of LLMs

From one word at a time to agents • 2014 → 2026

**The Transformer architecture is nine years old.
ChatGPT is almost four years old.**

June 2017 • November 2022



Beginning
of the
Journey

<2017

The problem

Translation, seq2seq

2018-19

Encoders vs. GPT

BERT reads, GPT writes

2021-22

RLHF

Creating assistants with LLMs

2024

Multimodality

Vision and memory

2025

MoE & agents

DeepSeek, tool calling

2017

Transformer

attention is all you need

2019-20

Scaling

GPT-3, 175B, few-shot

2023

The race

OpenAI, Google, Anthropic

2024-25

Reasoning

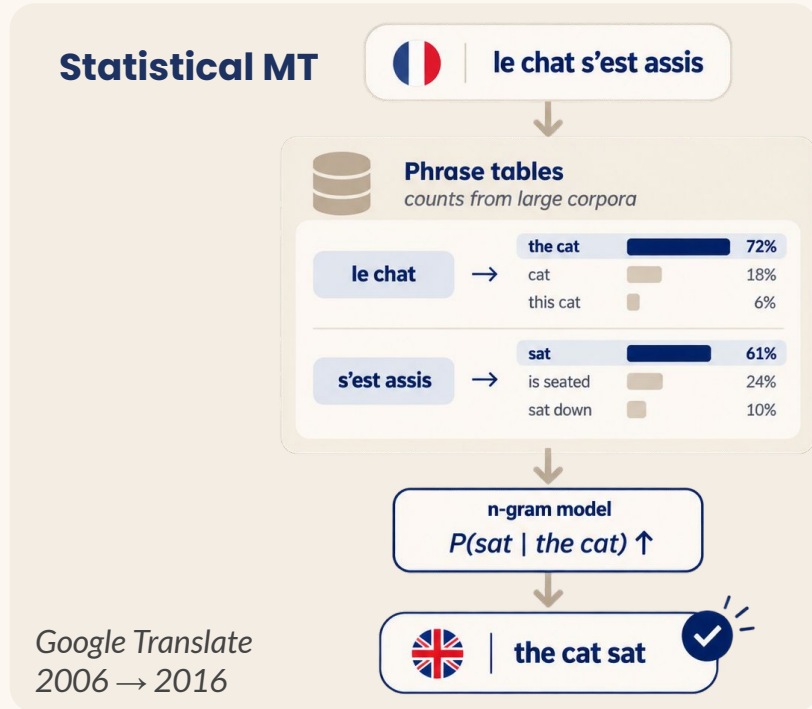
Spend tokens to think

2026

Latest advances

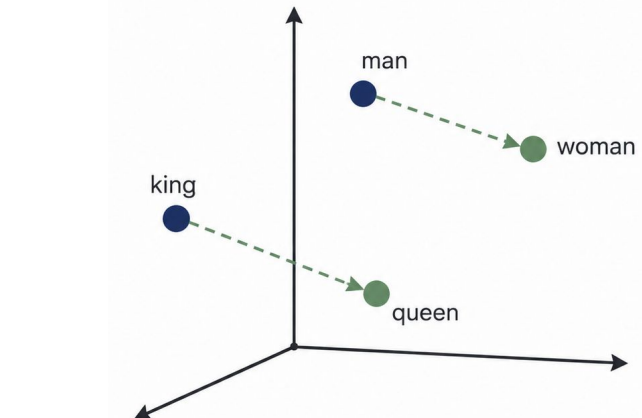
Headlines about AI

Before 2017: translation, by counting



Word2vec • 2013

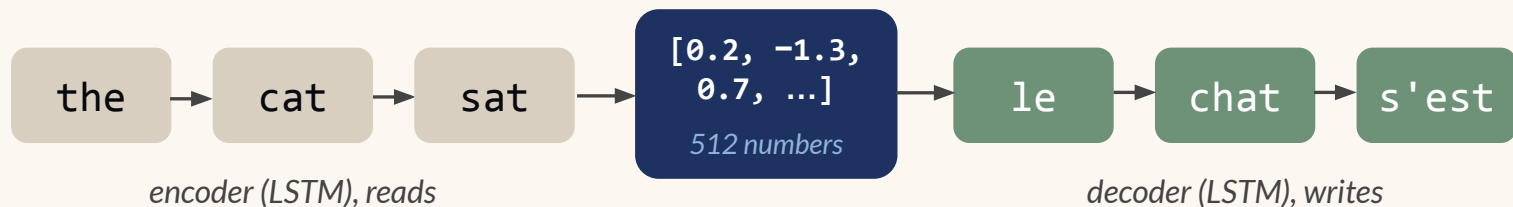
Mikolov et al. • Google



Similar meaning → nearby vectors

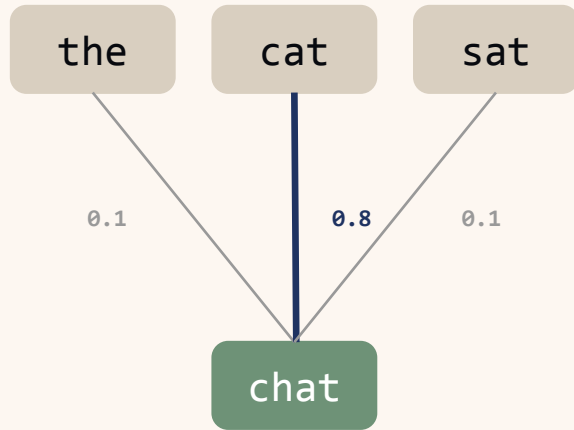
Meaning becomes arithmetic

2014: squeeze the sentence into one vector



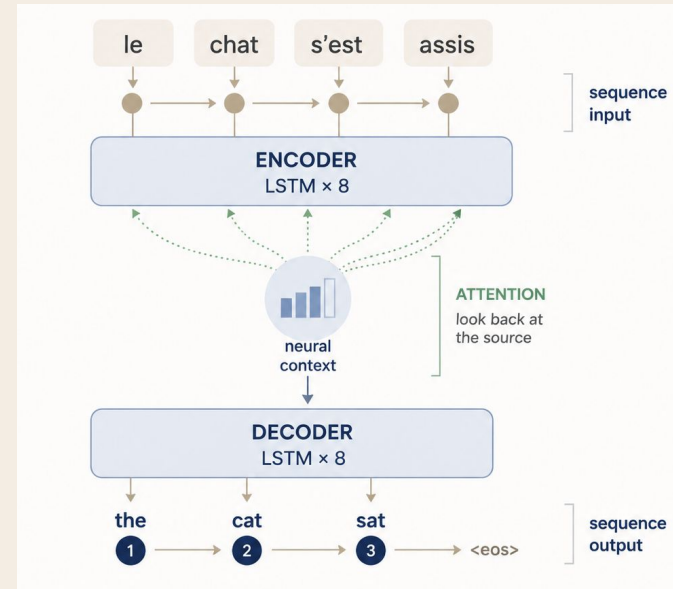
One fixed-size vector.
5 words or 50 words, same 512 numbers.

2015: attention, as a patch



The decoder weights every input word, every step.

GNMT • 2016 *Google Translate goes neural*



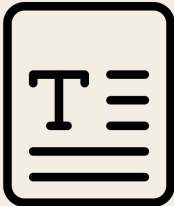
1

The diagram shows a sequence of six circular nodes arranged in a zig-zag pattern. The flow starts at the top left, moves right to the top right node, then down to the middle right node, then left to the middle left node, then down to the bottom left node, and finally right to the bottom right node. Each step is indicated by a straight line with an arrowhead pointing in the direction of the flow.

Sequential

Word n waits for word $n-1$
GPUs mostly idle
Vision went parallel in 2012

2



Long paths

Word 1 to word n crosses n steps
The signal fades
Long dependencies are lost

3

Cost

- Low training caps size and data
- Weeks per experiment
- No way to scale

2017: Attention Is All You Need

Attention Is All You Need

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Niki Parmar*
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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks in an encoder-decoder configuration. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.0 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

1 Introduction

arXiv:1706.03762v1 [cs.CL] 12 Jun 2017

Vaswani et al. • Google • June 2017

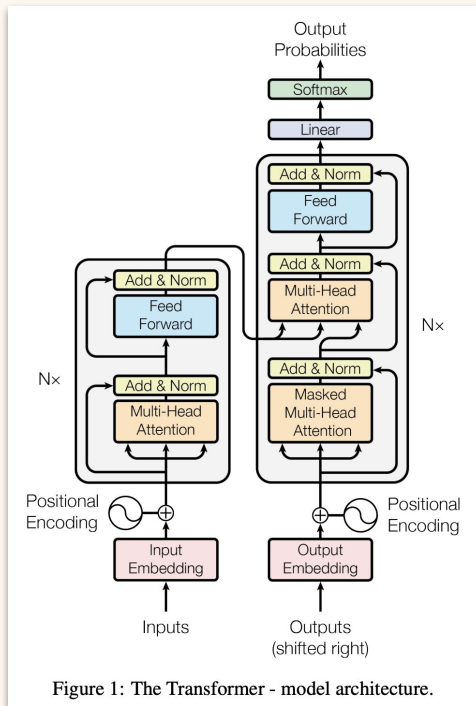


Figure 1: The Transformer - model architecture.

Encoder

Reads the input, all words at once

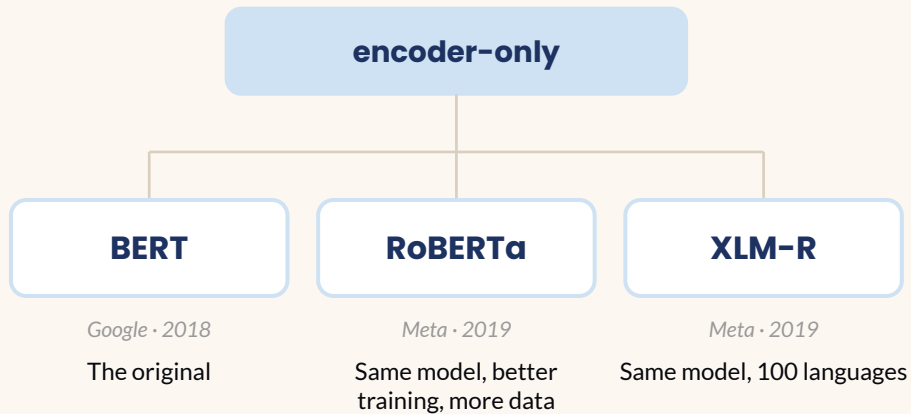
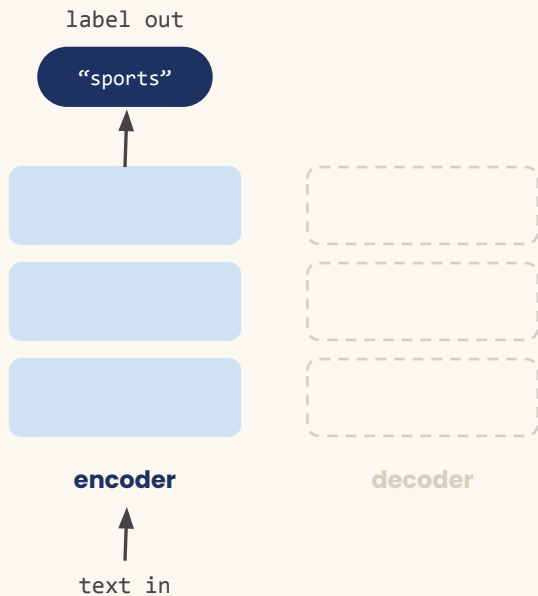
Decoder

Writes the output, one token at a time

3.5 days • 8 GPUs

BLEU 28.4 on WMT'14 En→De
built for translation

Encoders vs. GPT: models that read



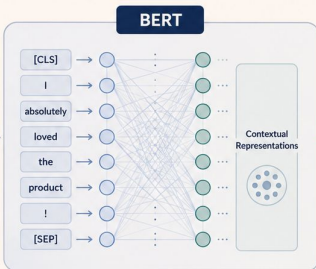
Encoders vs. GPT: models that read

Sentiment Analysis using BERT

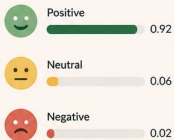
BERT understands the context of the text and predicts the overall sentiment.

INPUT TEXT

“
I absolutely loved
the product! The
quality is outstanding
and the customer
service was
excellent.”



SENTIMENT SCORES



PREDICTION



BERT is a pre-trained language model that captures deep contextual relationships by looking at the full sentence in both directions.

BERT
Bidirectional Encoder Representations from Transformers

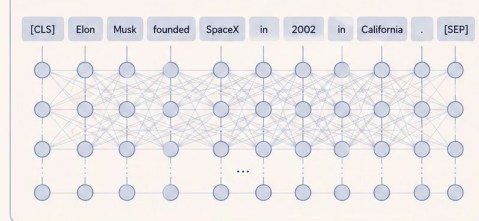
NAMED ENTITY RECOGNITION USING BERT

BERT understands the context of the sentence and identifies and classifies named entities.

EXAMPLE INPUT

Elon Musk founded SpaceX in 2002 in California .

BERT ENCODING



ENTITY PREDICTIONS

TOKEN	PREDICTED LABEL
[CLS]	O
Elon	B-PER
Musk	I-PER
founded	O
SpaceX	B-ORG
in	O
2002	B-DATE
in	O
California	B-LOC
.	O
[SEP]	O

ENTITY TYPES

PER Person ORG Organization LOC Location DATE Date / Time O Other

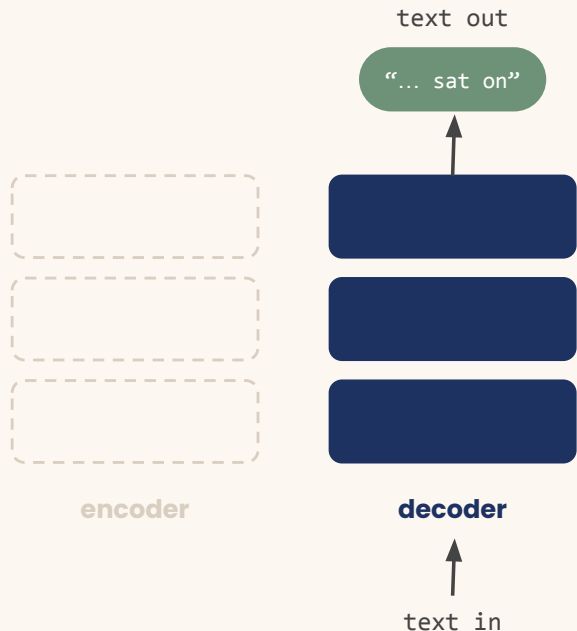
FINAL OUTPUT

Elon Musk founded SpaceX in 2002 in California .
PER PER O ORG O DATE O LOC O



BERT (Bidirectional Encoder Representations from Transformers) is a pre-trained language model that captures deep contextual relationships by looking at the full sentence in both directions.

Encoders vs. GPT: the contrarian bet



Alec Radford

Independent
No verified email

[Deep Learning](#) [Machine Learning](#)

[FOLLOW](#)

[GET MY OWN PROFILE](#)

Cited by	All	Since 2021
Citations	399773	373510
h-index	60	60
i10-index	71	71

Year	Citations
2019	~10,000
2020	~15,000
2021	~25,000
2022	~35,000
2023	~55,000
2024	~85,000
2025	~115,000
2026	~118,000

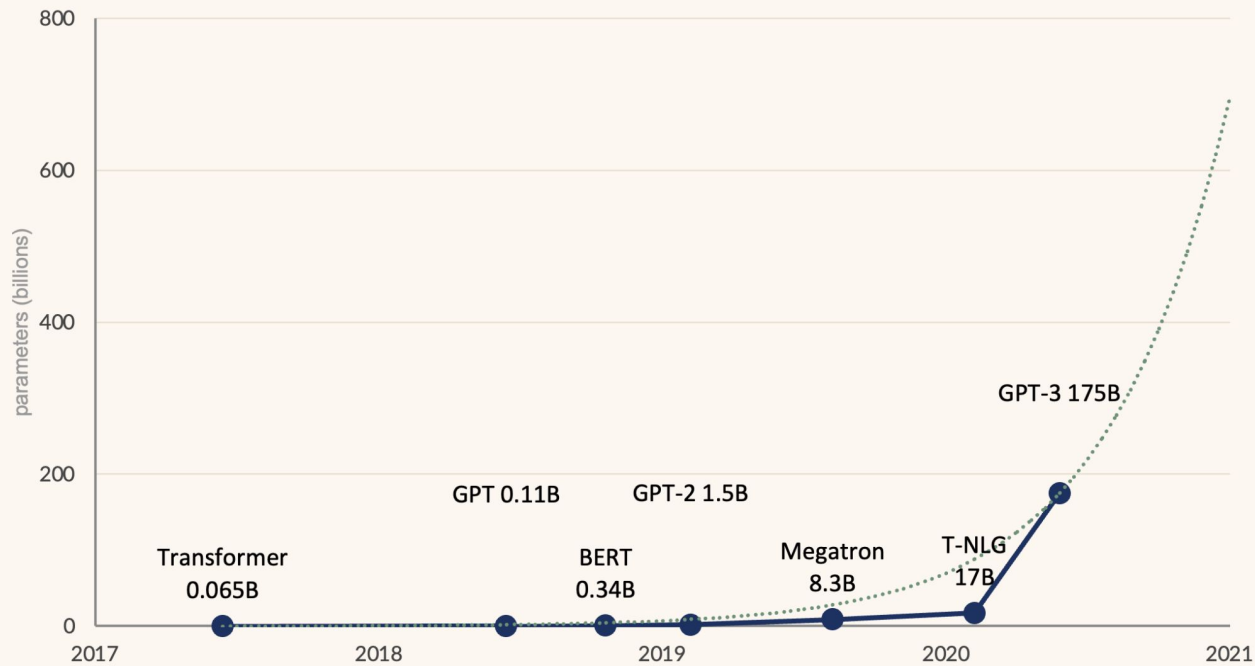
Co-authors

- Ilya Sutskever**
Co-Founder and Chief Scientist ...
- Soumith Chintala**

TITLE	CITED BY	YEAR
Language models are few-shot learners T Brown, B Mann, N Ryder, M Subbiah, JD Kaplan, P Dhariwal, ... Advances in neural information processing systems 33, 1877-1901	81168	2020
Learning Transferable Visual Models From Natural Language Supervision A Radford, JW Kim, C Hallacy, A Ramesh, G Goh, S Agarwal, G Sastry, ... https://cdn.openai.com/papers ...	71498	2021
Proximal policy optimization algorithms J Schulman, F Wolski, P Dhariwal, A Radford, O Klimov arXiv preprint arXiv:1707.06347	44457	2017
Language Models are Unsupervised Multitask Learners A Radford, J Wu, R Child, D Luan, D Amodei, I Sutskever Technical report, OpenAI	40461 *	2019
Gpt-4 technical report J Achiam, S Adler, S Agarwal, L Ahmad, I Akkaya, FL Aleman, D Almeida, ... arXiv preprint arXiv:2303.08774	27596	2023

Alec Radford · led GPT (2018) and GPT-2 (2019)

Scaling: NLP's Moore's law



~10× per year

• dense models

... ×10 per year

GPT-3: 100× GPT-2 in one year
570 GB of text · ~500B tokens

Scaling: the task lives in the prompt

prompt

A “whatpu” is a small, furry animal native to Tanzania. An example of a sentence that uses the word whatpu is:

We were traveling in Africa and we saw these very cute whatpus.

To do a “farduddle” means to jump up and down really fast. An example of a sentence that uses the word farduddle is:

completion

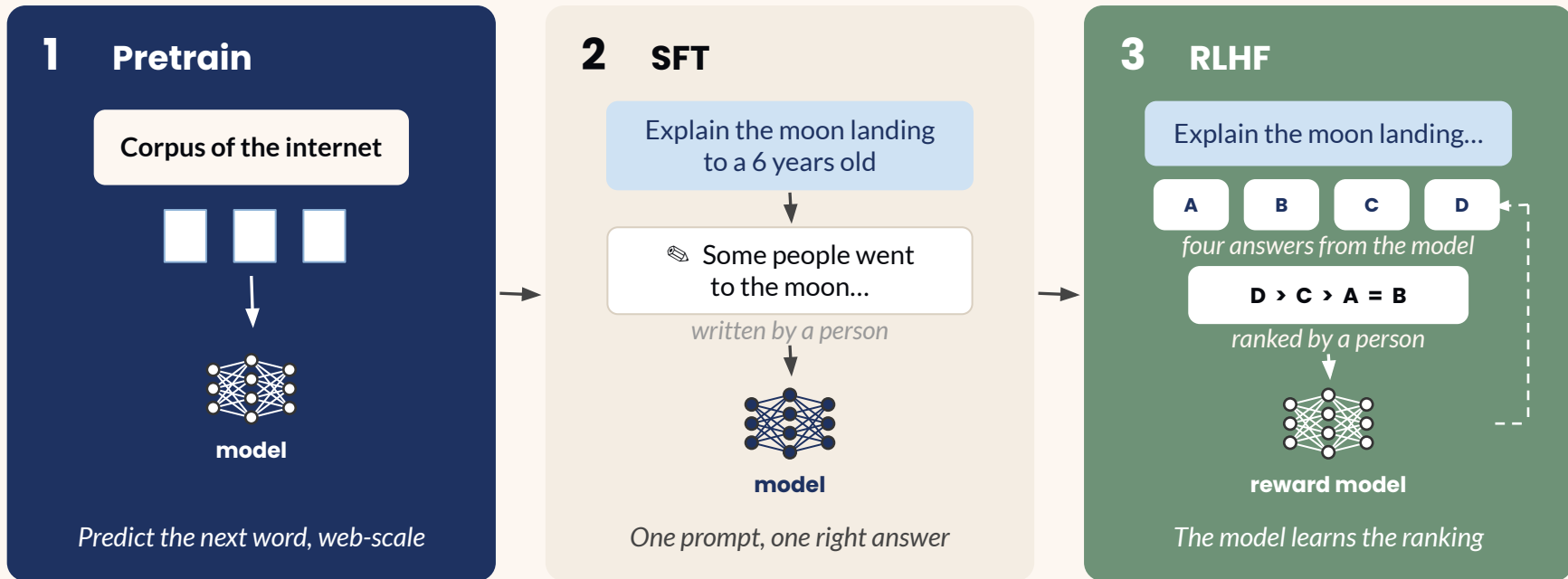
One day when I was playing tag with my little sister, she got really excited and she started doing these crazy farduddles.█

No retraining

One example of the shape, definition then sentence.

It produces another with a word that **does not exist**.

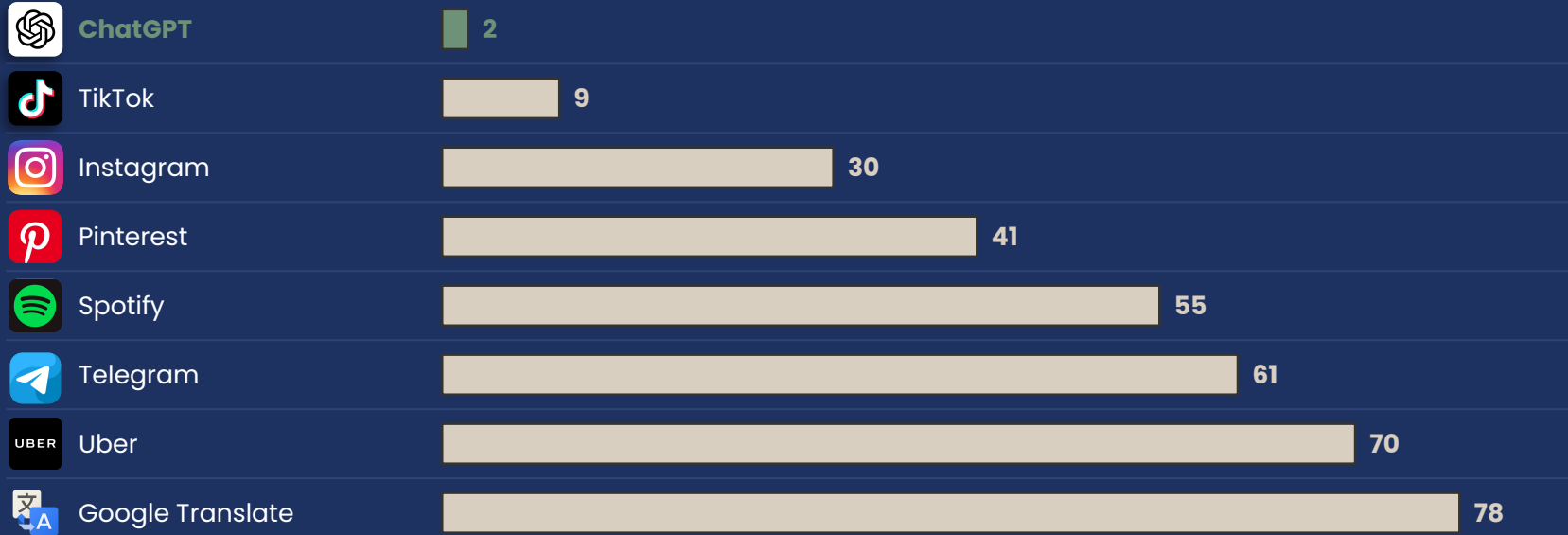
RLHF: teaching it to talk



Nov 2022

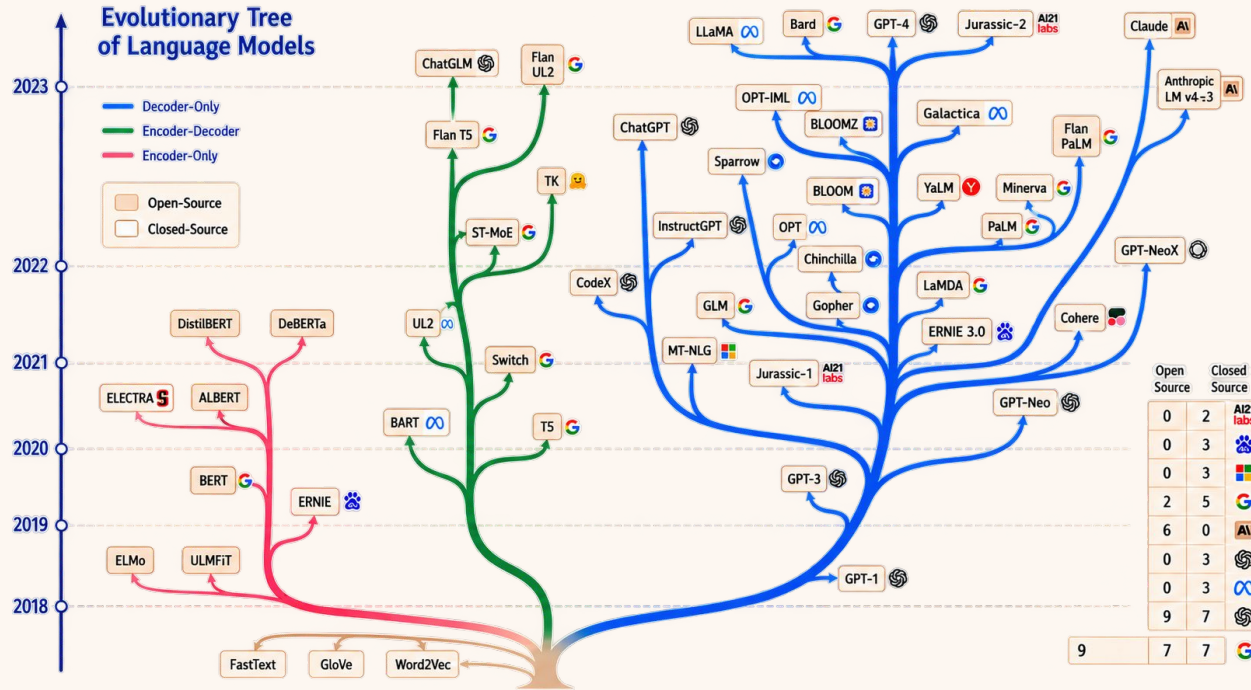
ChatGPT = InstructGPT + engineering

months to reach 100M monthly users



source: UBS, Feb 2023 (reported by Reuters / Yahoo Finance)

The race: everyone joins



Adapted from "Harnessing the Power of LLMs in Practice: A Survey on ChatGPT and Beyond", 2023.

The race: limitations of ChatGPT

What happened to Shell on December 12th 2017?



On December 12th, 2017, Shell made a significant announcement regarding its climate change strategy. Shell pledged to reduce its carbon footprint, setting ambitious targets to halve the net carbon footprint of its energy products by 2050 and to reduce it by around 20% by 2035. This move was part of Shell's broader efforts to align its business with the goals of the Paris Agreement, aiming to limit global warming to well below 2 degrees Celsius above pre-industrial levels.

Shell's commitment included a range of measures, such as investing in renewable energy, improving energy efficiency, and developing carbon capture and storage technologies. The company also emphasized its intention to regularly review and adjust its targets in response to technological advancements and policy changes.

This announcement was seen as a major step by one of the world's largest oil and gas companies to address climate change and transition towards a lower-carbon future. It reflected the growing pressure on energy companies to take more decisive action on environmental sustainability and reduce their impact on the planet.



Factual hallucination



United Nations
Climate Change



EN ▾



Shell Vows to Halve Carbon Footprint by 2050 to Achieve Paris Goals

28 November 2017

Share the article



Announcement



Correct date of the announcement. Source: United Nations Climate Change

The race: limitations of ChatGPT



Summarize the essay "Billionaires Build, by Paul Graham"



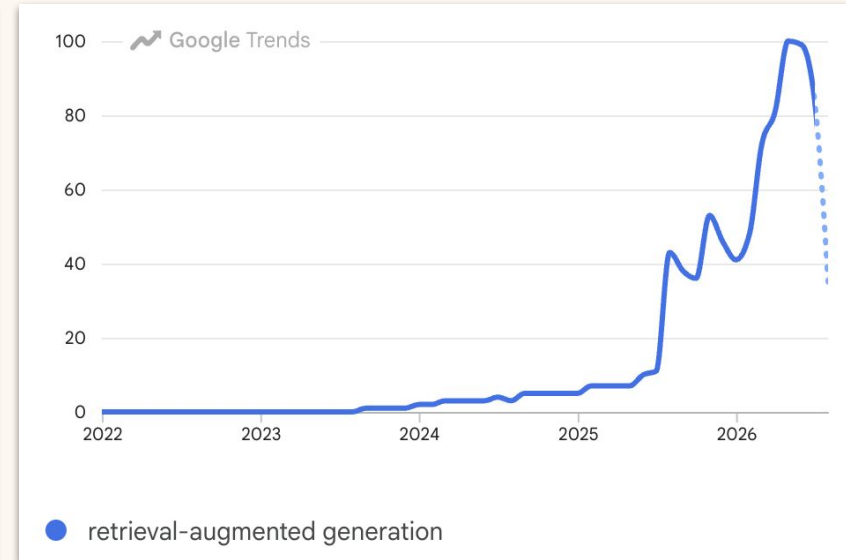
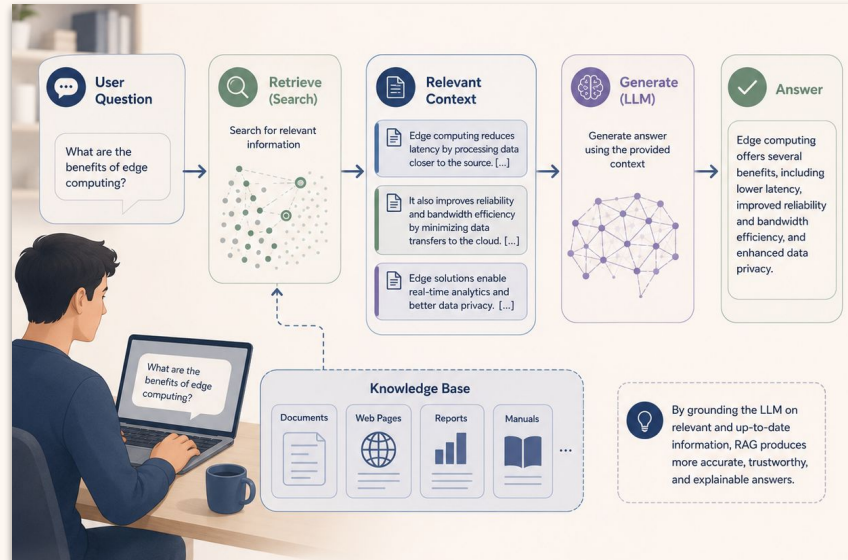
I'm sorry, but as of my last training cutoff in September 2021, there isn't an essay titled "Billionaires Build" by Paul Graham that I can summarize. Paul Graham is an entrepreneur and co-founder of Y Combinator, and he's known for his essays about startups, technology, and society. However, if this essay was published after September 2021, I can't provide a summary as my training data doesn't include it. If you have any key points or ideas from the essay, I'd be more than happy to discuss and explain them.



Knowledge cut-off

The race: your data in the prompt

RAG · Retrieval-Augmented Generation



Multimodality: vision and memory

1,000,000

Tokens in one prompt · Gemini 1.5 Pro · Feb 2024

1 h

of video

11 h

of audio

30k

lines of code

700k

words

402 pages of Apollo 11 transcripts, one prompt.

4k

2022 · ChatGPT

32k

2023 · GPT-4

128k

2023 · GPT-4 Turbo

200k

2023 · Claude 2.1

1M

2024 · Gemini 1.5

Example of GPT-4 visual input:

User What is funny about this image? Describe it panel by panel.



Source: <https://www.reddit.com/r/hmmm/comments/ubab5v/hmmm/>

sees images ✓
generates images ✗

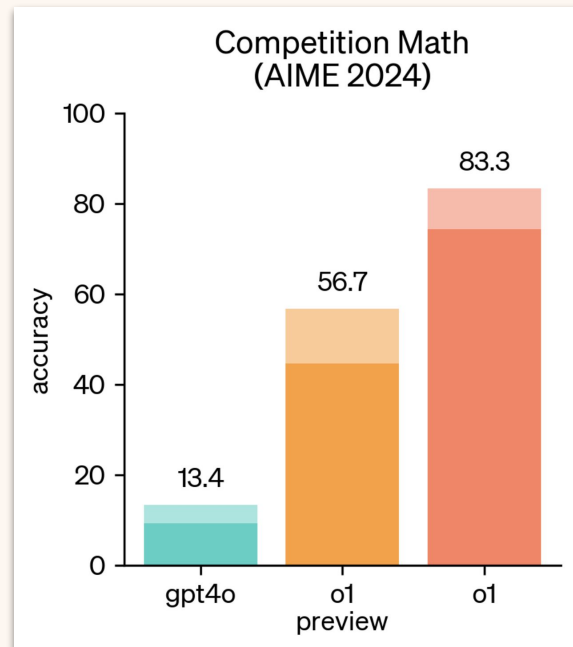
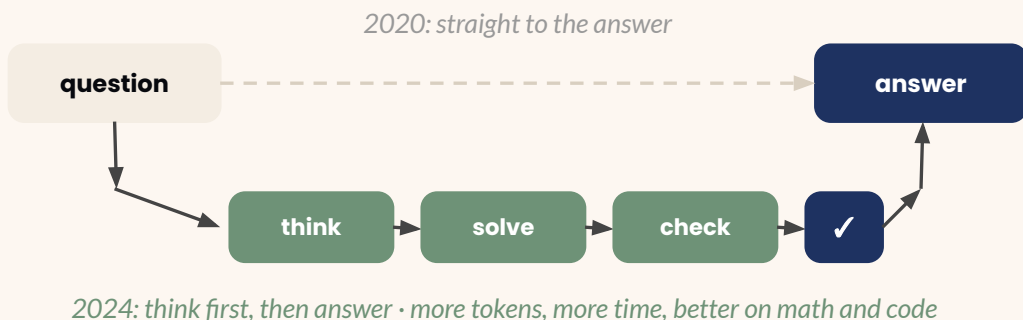
sources: *blog.google*, Gemini 1.5 launch (Feb 2024) · GPT-4 technical report (OpenAI, Mar 2023)

Reasoning: spend tokens to think

13% → 83%

GPT-4o vs o1 on the AIME (the IMO qualifying exam) · Sep 2024

Same lab, same year. The difference is thinking before answering.



Source: OpenAI's blog · Learning to reason with LLMs, 2024.

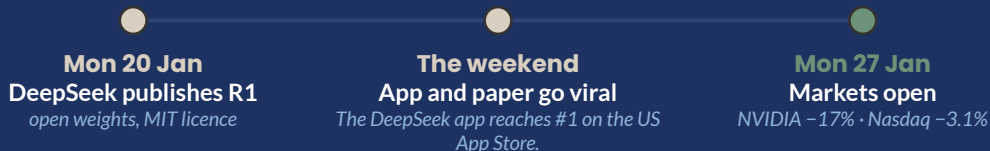


MoE & agents: the DeepSeek bomb

-\$589B

Nvidia became the company with the biggest single-day loss ever.

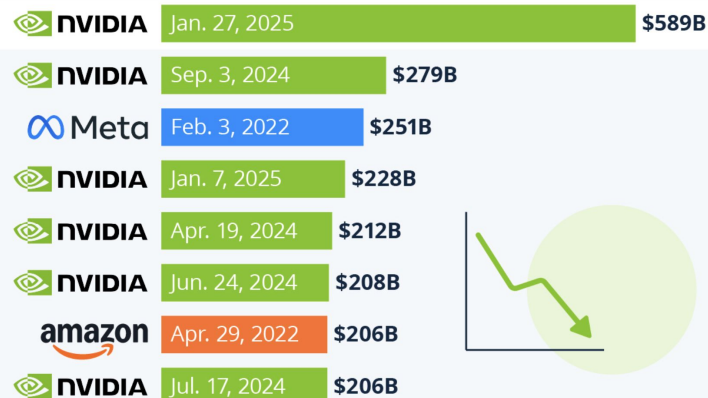
One week earlier, NVIDIA had just overtaken Apple as the most valuable company in the world.



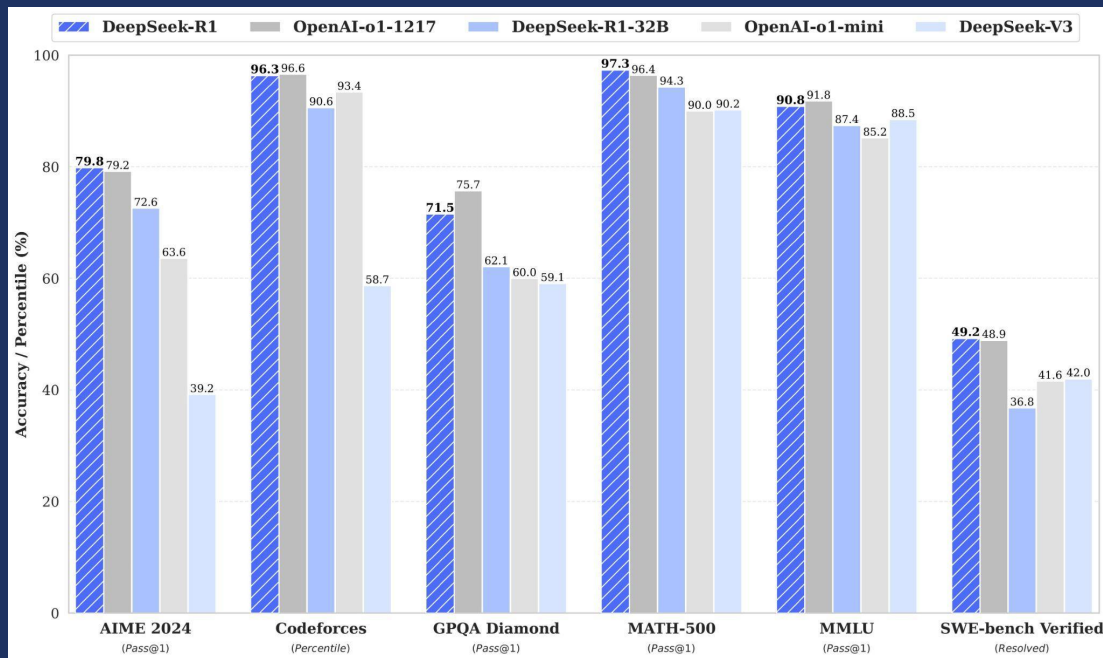
Sources: CNBC, Forbes, Statista (27-28 Jan 2025)

Nvidia Sheds Nearly \$600 Billion in Historic Single-Day Drop

Largest single-day drops in market capitalization of U.S. public companies



MoE & agents: why it happened



R1 vs o1 · AIME 79.8 / 79.2 · GPQA 71.5 / 75.7 · Codeforces 96.3 / 96.6 · SWE-bench 49.2 / 48.9

o1-level

A tie with the best closed model in the world, on math, code and science.

~10% of the cost

\$5.6M declared for V3 vs >\$100M estimated for GPT-4.

open weights, MIT

Anyone can download it, run it, fine-tune it, sell it



O1

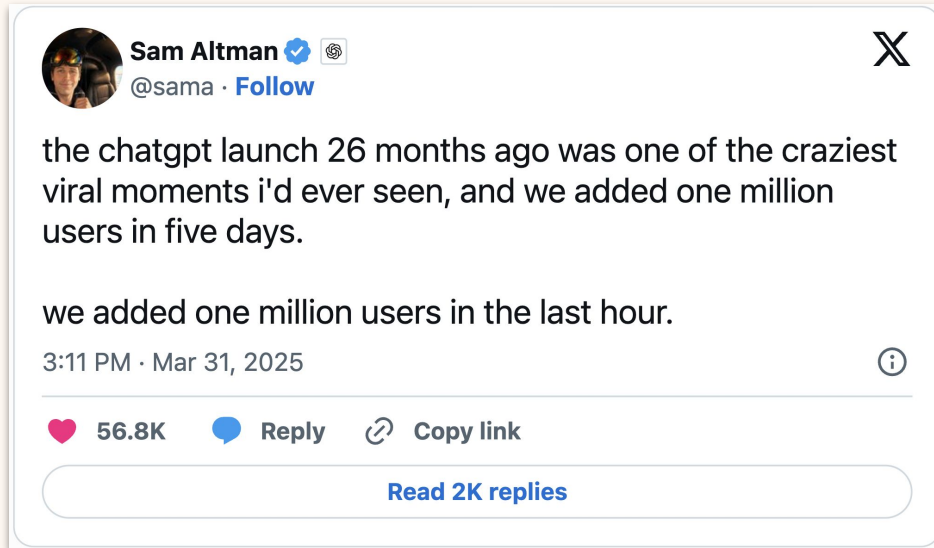
Dense model. All parameters are active.



DeepSeek

Mixture of Experts (MoE). 2 out of 8 are active.

MoE & agents: one model for pixels



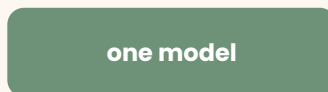
2024

*sees images*

diffusion

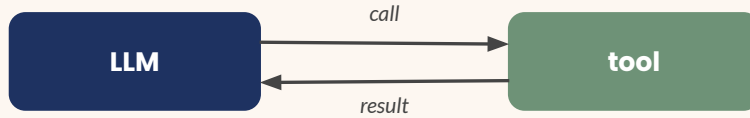
makes images

2025

*sees and makes*

Ghibli style GPT 4o image generation

MoE & agents: tool calling



Call a tool, read the result, decide the next step.

Run locally



Ollama

Any open model on your laptop

Code



**LangChain ·
LangGraph · CrewAI**

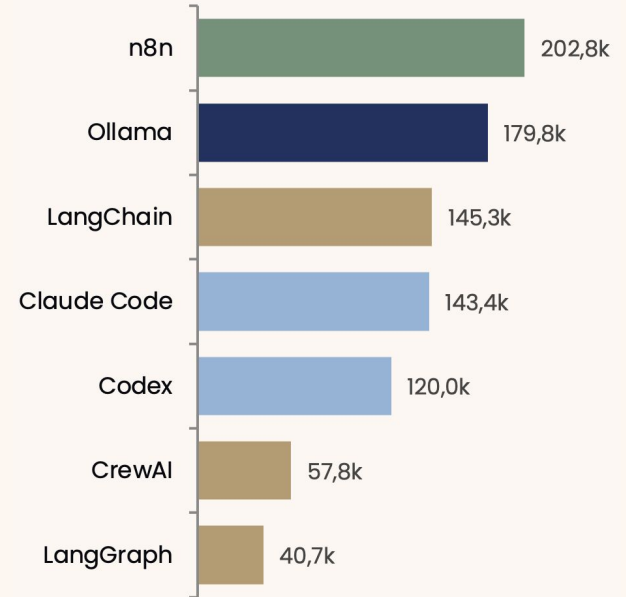
Chains, graphs, crews of agents

No code



n8n · Langflow

Drag boxes, wire them



GitHub stars, 30 Aug 2026

MoE & agents: a USB port for tools

6 months

Nov 2024

Anthropic publishes MCP
Open spec

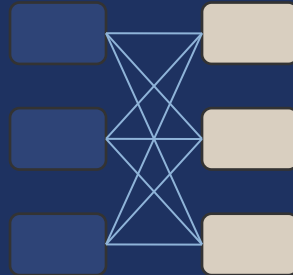
Mar 2025

OpenAI adopts it
ChatGPT, Agents SDK

Apr 2025

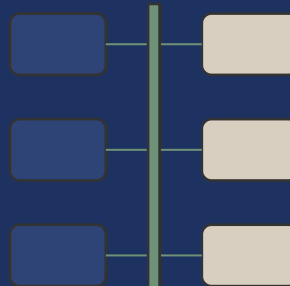
Google adopts it
and proposes A2A, agent ↔ agent

before



$N \times M$ integrations

after



MCP

$N + M$

12 MCP Servers You Can Use in 2025 ByteByteGo

1 File System Server Gives the LLM direct access to the local file system to read, write, create directories	2 GitHub MCP Server Connects Claude to GitHub repos and allows file updates, code searching	3 Slack MCP Server MCP Server for Slack API, enabling Claude to interact with Slack workspaces
4 Google Maps MCP Server MCP Server for Google Maps API	5 Docker MCP Server Integrate with Docker to manage containers, images, volumes, and networks	6 Brave MCP Server Web and local search using Brave's Search API
7 PostgreSQL MCP Server A MCP Server that enables LLM to inspect database schemas and execute read-only queries	8 Google Drive MCP Server MCP Server integrates with Google Drive to allow reading and searching over files	9 Redis MCP Server MCP Server that provides access to Redis databases
10 Notion MCP Server This project implements an MCP Server for the Notion API	11 Stripe MCP Server MCP Server to interact with the Stripe API	12 Perplexity MCP Server An MCP Server that connects to Perplexity's Sonar API for real-time search

Source: ByteByteGo, 2025.

MoE & agents: memory and skills

Memory

Personalization

Custom instructions

[Save](#)

Give ChatGPT extra instructions and context for all chats on this host. [Learn more](#)

Add your custom instructions...

Memory

Configure how local memories are collected, retained, and consolidated on this computer. [Learn more](#)

Enable local memories

Create memories from chats on this computer and use them to personalize future chats on this computer



Allow local memory generation from tool-assisted chats

Generate memories from chats that used MCP tools or web search



Delete local memories

Delete all memories stored locally on this computer

[Delete](#)

Skills

Plugins

[Browse directory](#)[Add](#)

Manage plugins, skills, and MCPs

[Plugins 10](#)[Apps 5](#)[MCPs 5](#)[Skills 24](#)

Academic Research Suite

ARS workflows plus Claude-style ars-* aliases

Personal



Google Agents CLI Adk Code

This skill should be used when the user wants to "write agent code", "build a...

Personal



Google Agents CLI Deploy

This skill should be used when the user wants to "deploy an agent", "deploy...

Personal



Google Agents CLI Eval

This skill should be used when the user wants to "run an evaluation", "evalua...

Personal



Google Agents CLI Observability

This skill should be used when the user wants to "set up tracing", "monitor m...

Personal



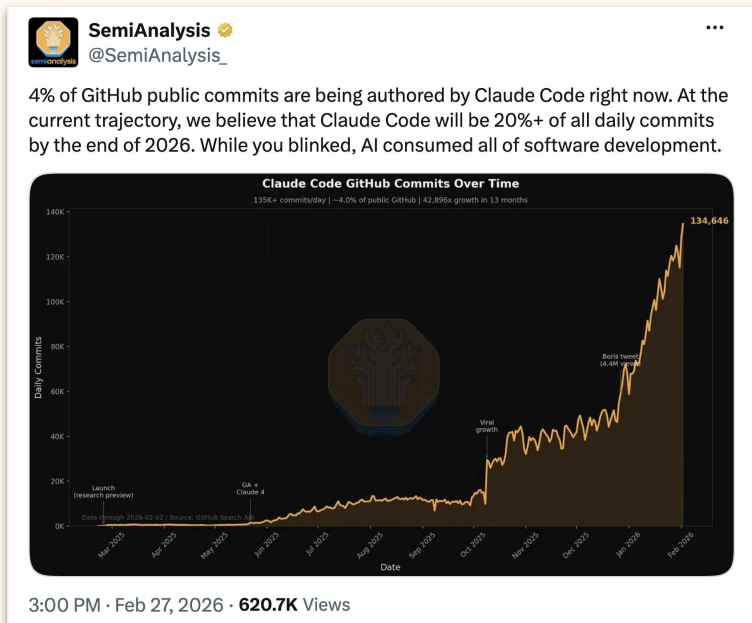
Google Agents CLI Publish

This skill should be used when the user wants to "publish an agent", "publish...

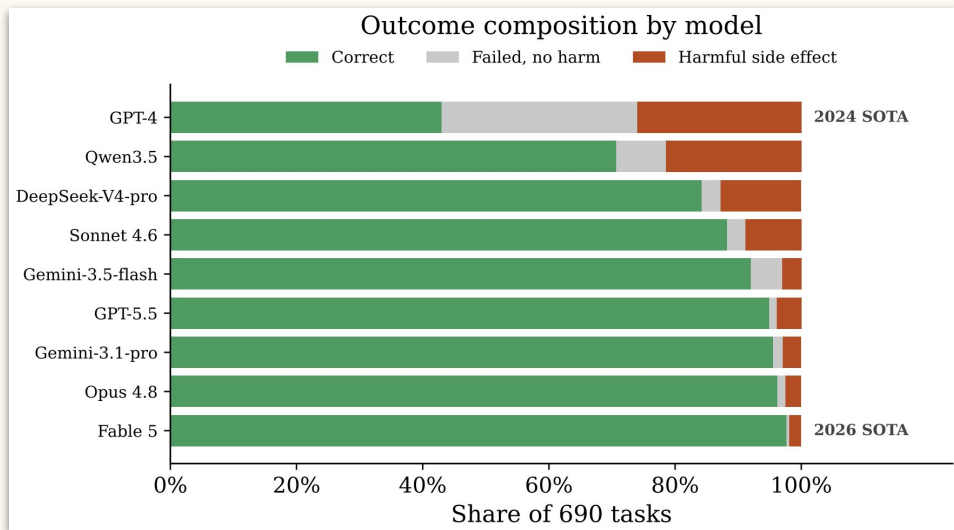
Personal



Latest advances: the headlines (1/2)



x.com/SemiAnalysis_/status/2027443723362042308



WorkBench Revisited, 2026. Source: arxiv.org/pdf/2606.13715

Latest advances: the headlines (2/2)

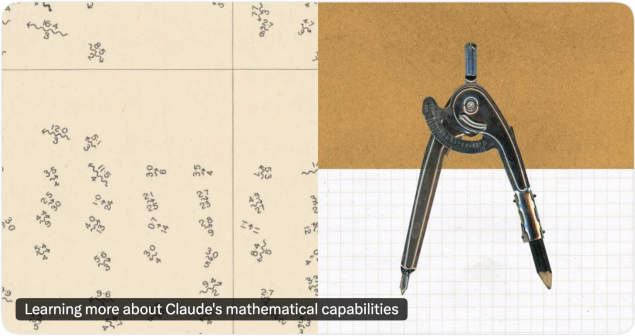

Anthropic


 @AnthropicAI

We asked an unreleased research version of Claude to take a stab at the Riemann hypothesis.

It didn't solve it, but it did make strides on a related problem: it increased the lower bound for the fraction of zeros of the Riemann zeta function that satisfy the hypothesis from 41.6% to 67.2%.

anthropic.com/research/riema...



Learning more about Claude's mathematical capabilities

From anthropic.com

2:28 PM · Aug 10, 2026 · **10.2M** Views

x.com/AnthropicAI/status/2086867246073401655


Hugging Face

[← Back to Articles](#)

Anatomy of a Frontier Lab Agent Intrusion: A Technical Timeline of the July 2026 Incident

Published July 27, 2026

[Update on GitHub](#)

▲ Upvote **482**  +476


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[Xc1D](#) Follow


raphael g
[raphael-gl](#) Follow


Christophe Rannou
[chris-rannou](#) Follow

A companion technical writeup to our [incident disclosure](#). This post walks through how the intrusion actually worked: the two initial-access vectors, how the agent pivoted and moved laterally, representative examples of the commands that were run and how we investigated with GLM 5.2 (an open-source model). Live credentials, internal hostnames, and specific indicators have been redacted or genericized, while the techniques are described exactly as observed by Hugging Face.

huggingface.co/blog/agent-intrusion-technical-timeline

Clone me!!!!

The screenshot shows a Google Colab notebook interface. At the top, the title bar says 'lesson05.ipynb' with a star icon. Below it are tabs for 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. A search bar contains 'Commands'. To the right are icons for chat, settings, a 'Share' button, and a user profile. Below the tabs is a toolbar with 'Q Commands', '+ Code', '+ Text', 'Run all', and 'Connect T4'. On the left is a sidebar with icons for file explorer, search, and other functions. The main content area has a title 'Before the Transformer: squeeze, then attend' and a paragraph: 'Today's lecture told a story that starts with a problem: **translation**. This notebook lets you touch the two ideas that opened it, with real data and a real metric.' Below this is a table with four columns: 'Section', 'Year', 'The idea', and 'The slide it comes from'. The table has two rows: '1 · One vector' (2014) and '2 · Attention' (2015). Below the table is a paragraph: 'We stop right before 2017, on purpose. On Wednesday we build the Transformer piece by piece; today you will feel **why** it had to exist.' This is followed by a note: 'Runs on a free Colab GPU in about five minutes. Runtime → Change runtime type → GPU.' Below this is a section titled 'Learning objectives' with a list of four items: 1. Train a 2014-style encoder-decoder translator, reusing the Architecture class from lesson 3b. 2. Measure the fixed-vector bottleneck: quality as a function of sentence length. 3. Add attention and watch the same measurement change. 4. Read an alignment map: where the model looked while writing each word.

Section	Year	The idea	The slide it comes from
1 · One vector	2014	squeeze the whole sentence into one vector, then write the translation from it	squeeze the sentence into one vector
2 · Attention	2015	let the decoder look back at every input word, with weights	attention, as a patch

Runs on a free Colab GPU in about five minutes. Runtime → Change runtime type → GPU.

Learning objectives

1. **Train** a 2014-style encoder-decoder translator, reusing the `Architecture` class from lesson 3b.
2. **Measure** the fixed-vector bottleneck: quality as a function of sentence length.
3. **Add** attention and watch the same measurement change.
4. **Read** an alignment map: where the model looked while writing each word.